```
clc;
clear;
close all;

% DESIGN AND ANALYSIS OF FIR DIGITAL FILTERS
% Using Fourier Series and Windowing Techniques

Fs = 8000;          % Sampling frequency (Hz)
Fc = 1800;          % Cutoff frequency (Hz)
N = 50;             % Filter order
n = 0:N;
alpha = N/2;

% Step 1: Ideal FIR low-pass filter using Fourier Series
Wc = 2*pi*Fc/Fs;
m = n - alpha;

hd = zeros(1,N+1);

for k = 1:length(n)
    if m(k) == 0
        hd(k) = Wc/pi;
    else
        hd(k) = sin(Wc*m(k))/(pi*m(k));
    end
end

% Step 2: Window functions
w_rect = ones(1,N+1);

w_hamm = 0.54 - 0.46*cos(2*pi*n/N);

w_hann = 0.5 - 0.5*cos(2*pi*n/N);

w_black = 0.42 - 0.5*cos(2*pi*n/N) ...
          + 0.08*cos(4*pi*n/N);

% Step 3: Apply windows to ideal impulse response
h_rect = hd .* w_rect;
h_hamm = hd .* w_hamm;
h_hann = hd .* w_hann;
h_black = hd .* w_black;

% Step 4: Calculate frequency responses
[H1,f] = freqz(h_rect,1,8192,Fs);
H2 = freqz(h_hamm,1,8192,Fs);
H3 = freqz(h_hann,1,8192,Fs);
H4 = freqz(h_black,1,8192,Fs);

% Step 5: Convert magnitude into dB
M1 = 20*log10(abs(H1)+1e-8);
M2 = 20*log10(abs(H2)+1e-8);
```

```
M3 = 20*log10(abs(H3)+1e-8);
M4 = 20*log10(abs(H4)+1e-8);

% Step 6: Create figure matching reference image
figure('Color','w','Position',[100 50 900 700]);

% TOP GRAPH: Magnitude Response
subplot(2,1,1);

plot(f,M1,'k','LineWidth',1.2);
hold on;
plot(f,M2,'b','LineWidth',1.2);
plot(f,M3,'g','LineWidth',1.2);
plot(f,M4,'r','LineWidth',1.2);

grid on;
xlim([0 4000]);
ylim([-100 10]);

xticks(0:500:4000);
yticks(-100:20:0);

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('FIR Low-Pass Filter Magnitude Response (N = 50)');

legend('Rectangular','Hamming','Hanning','Blackman', ...
       'Location','northeast');

% BOTTOM GRAPH: Hamming Phase Response
subplot(2,1,2);

phase = angle(H2)*180/pi;

plot(f,phase,'b','LineWidth',1.2);

grid on;
xlim([0 4000]);
ylim([-200 200]);

xticks(0:500:4000);
yticks(-200:100:200);

xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Linear Phase Response (Hamming Window Example)');

% Overall title
sgtitle('FIR Windowing Techniques Comparison');

% Display filter coefficients
disp('Hamming Window FIR Coefficients:');
disp(h_hamm);
```

```
Hamming Window FIR Coefficients:
  Columns 1 through 7

   -0.0007    0.0007    0.0012   -0.0005   -0.0020    0.0000    0.0034

  Columns 8 through 14

    0.0013   -0.0049   -0.0040    0.0060    0.0083   -0.0057   -0.0144

  Columns 15 through 21

    0.0028    0.0217    0.0041   -0.0298   -0.0172    0.0376    0.0411

  Columns 22 through 28

   -0.0441   -0.0915    0.0485    0.3133    0.4500    0.3133    0.0485

  Columns 29 through 35

   -0.0915   -0.0441    0.0411    0.0376   -0.0172   -0.0298    0.0041

  Columns 36 through 42

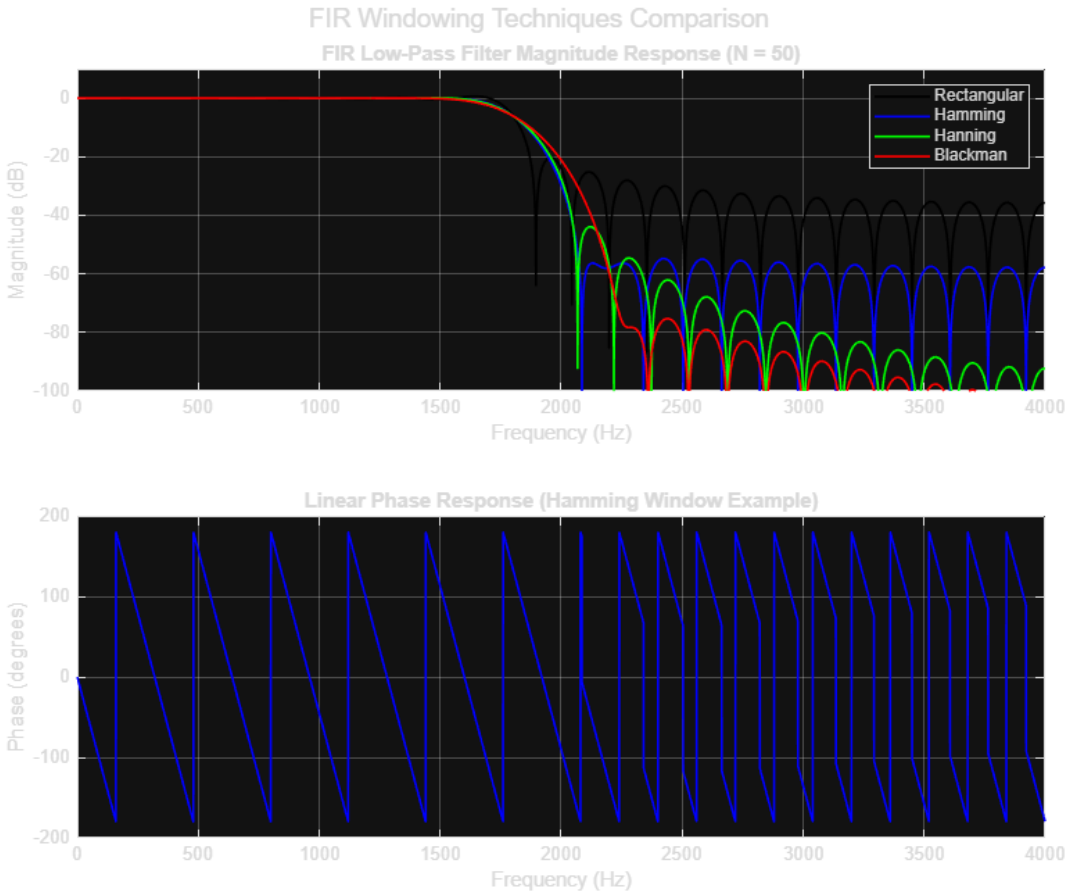
    0.0217    0.0028   -0.0144   -0.0057    0.0083    0.0060   -0.0040

  Columns 43 through 49

   -0.0049    0.0013    0.0034    0.0000   -0.0020   -0.0005    0.0012

  Columns 50 through 51

    0.0007   -0.0007
```



Published with MATLAB® R2026b